🎯 Objective

To understand cybersecurity fundamentals, analyze different cyberattacks, learn the working of SOC, and gain knowledge of advanced security technologies like SIEM, XDR, and SOAR along with Git basics.

🛡 Cybersecurity

Cybersecurity is the practice of protecting systems, networks, applications, and data from cyber threats, unauthorized access, and malicious activities.

In today's digital world, organizations store sensitive information such as personal data, financial records, and business data. Cybersecurity ensures that this information is protected from attacks.

🔹 Why Cybersecurity is Important

- Prevents data breaches
- Protects user privacy
- Ensures safe online transactions
- Maintains trust in digital systems
- Prevents financial loss

🔹 Types of Assets Protected

Cybersecurity focuses on protecting:

- **Hardware** → Computers, servers
- **Software** → Applications, operating systems
- **Data** → Personal, financial, confidential data
- **Networks** → Internet, internal communication

⚠ Types of Cyber Attacks

Cyber attacks are malicious attempts to access, damage, or disrupt systems, networks, or data. Understanding these attacks helps in detecting and preventing them effectively.

🔹 1. Phishing Attack

Phishing is a **social engineering attack** where attackers trick users into revealing sensitive information such as passwords, OTPs, or bank details.

🔹 How it works:

- Attacker sends fake email/message
- Appears like a trusted source (bank, company)
- Contains malicious link
- User enters credentials
- Attacker steals data

🔹 Example:

Fake email saying:
👉 "Your bank account will be blocked. Click here to verify."

🔹 Prevention:

- Check URL carefully
- Avoid clicking unknown links
- Verify sender email
- Do not share sensitive data

🔗 Identifying Phishing Links

Phishing links are malicious URLs designed to trick users into revealing sensitive information such as login credentials, personal data, or financial details. These links often appear legitimate, so careful verification is required before clicking.

🔍 1. URL Structure Analysis

The URL is the most important factor in identifying a fake link.

- Look for spelling mistakes
  - Example: amaz0n.com instead of amazon.com
- Check for extra words
  - Example: secure-login, verify-account
- Avoid unusual or complex URLs

🌐 2. Domain Name Verification

Focus on the main domain name (the part before .com, .org, etc.).

- Example:
  - https://amazon.com ✔ (legitimate)
  - https://amazon-login.xyz ✕ (fake)
  - https://amazon.com.fake-site.xyz ✕

👉 Always identify the **actual domain**, not just the visible name.

🔒 3. HTTPS and Security Indicators

- HTTPS indicates encrypted communication
- However, phishing websites can also use HTTPS

👉 HTTPS alone does not guarantee safety

🖱 4. Hover to Inspect Link

Before clicking a link:

- Hover the mouse over the link
- Check the actual URL displayed

👉 If the visible link and actual link are different, it is suspicious

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✉ 5. Sender Verification

If the link is received via email or message:

- Check the sender's email address
- Look for unusual domains
  - Example: support@amaz0n-security.com ❌
- Be cautious of unknown senders

⚠ 6. Urgency and Suspicious Language

Phishing messages often create panic or urgency:

- "Your account will be blocked!"
- "Click immediately to verify!"

👉 This is used to force quick action without thinking

📎 7. Shortened Links

Shortened links hide the real destination:

- Examples: bit.ly, tinyurl.com

👉 Avoid clicking unless verified

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8. Website Content Verification

After opening a link, check:

- Poor design or layout
- Spelling or grammar mistakes
- Fake login pages
- Requests for sensitive information

🔐 9. Security Indicators

- Missing contact details
- No privacy policy
- Fake or copied logos

🔍 10. Manual Verification

- Type website URL manually in browser
- Do not rely on links from emails
- Use official sources

🎯 Quick Checklist

Before clicking any link:

- ✔ Is the domain correct?
- ✔ Any spelling mistakes?
- ✔ Is the sender trusted?
- ✔ Is the message urgent or suspicious?
- ✔ Is the link shortened?

👉 If any doubt → Do not click

🔹 2. Malware Attack

Malware is **malicious software** designed to damage systems or steal data.

🔹 Types of Malware:

- **Virus** → attaches to files
- **Worm** → spreads automatically
- **Trojan** → looks like legitimate software
- **Ransomware** → locks data and demands payment

🔹 Example:

Downloading a fake software that installs a virus

🔹 Impact:

- Data theft
- System damage
- Unauthorized access

🔹 3. Brute Force Attack

A brute force attack involves trying multiple username-password combinations until the correct one is found.

🔹 How it works:

- Automated tools try thousands of passwords
- Eventually finds correct one

🔹 Example:

Trying all possible passwords for a login system

🔹 Prevention:

- Strong passwords
- Multi-factor authentication (MFA)
- Account lock after failed attempts

🔷 4. Denial of Service (DoS) Attack

A DoS attack aims to make a system or service unavailable by overwhelming it with excessive traffic.

🔹 How it works:

- Attacker sends large number of requests
- Server becomes overloaded
- Real users cannot access

🔹 Example:

Website crash due to heavy fake traffic

🔹 Advanced form:

👉 DDoS (Distributed DoS)

Multiple systems attack simultaneously

🔹 Prevention:

- Firewalls
- Traffic filtering
- Load balancing

🔹 5. Man-in-the-Middle (MitM) Attack

In this attack, the attacker intercepts communication between two parties without their knowledge.

🔹 How it works:

- Attacker positions between user and server
- Captures or modifies data

🔹 Example:

Using public Wi-Fi to steal login credentials

🔹 Prevention:

- Use HTTPS
- Avoid public Wi-Fi
- Use VPN

🔹 6. Ransomware Attack

Ransomware is a type of malware that encrypts files and demands payment to restore access.

🔹 Example:

System shows message:
👉 “Pay money to unlock your files”

🔹 Prevention:

- Regular backups
- Avoid suspicious downloads
- Updated antivirus

🔷 7. SQL Injection

An attack where malicious SQL queries are inserted into input fields to access or manipulate database data.

🔹 Example:

Login form exploited to access database

🔹 Prevention:

- Input validation
- Secure coding practices

🔷 8. Insider Threat

Attack from inside the organization by employees or trusted users.

◆ Example:

Employee leaking confidential data

◈ 9. Zero-Day Attack

An attack that exploits unknown vulnerabilities before they are fixed.

📎 Summary Table

Attack Type	Purpose
Phishing	Steal credentials
Malware	Damage or steal data
Brute Force	Crack passwords
DoS/DDoS	Disrupt services
MitM	Intercept communication
Ransomware	Demand money
SQL Injection	Access database

🔐 CIA Triad (Core of Cybersecurity)

The CIA Triad is the **foundation of cybersecurity**.
It consists of three main principles:

👉 **Confidentiality, Integrity, Availability**

◈ 1. Confidentiality

Confidentiality ensures that sensitive information is **only accessible to authorized users**.

◆ Example:

- Password-protected accounts
- Private emails
- Bank details

◆ Methods to maintain confidentiality:

- Encryption
- Authentication (login systems)
- Access control

◆ If confidentiality fails:

👉 Data leaks or unauthorized access occurs

◈ 2. Integrity

Integrity ensures that data remains **accurate, consistent, and unmodified**.

◆ Example:

- Bank transaction data should not be altered
- Exam results should remain unchanged

◆ Methods to maintain integrity:

- Hashing
- Data validation
- Access control

◆ If integrity fails:

👉 Data can be tampered or corrupted

◈ 3. Availability

Availability ensures that systems and data are **accessible when needed**.

◆ Example:

- Websites should be accessible 24/7
- ATM systems should work anytime

◆ Methods to maintain availability:

- Backup systems
- Load balancing
- Protection against DoS attacks

◆ If availability fails:

👉 Systems become unavailable (downtime)

🔗 Relationship Between CIA Triad

All three work together:

- Confidentiality → Protects access
- Integrity → Protects accuracy
- Availability → Ensures access

👉 If one fails → system security is compromised

🎯 Simple Example

👉 Online Banking System:

- Confidentiality → Only you can access your account
- Integrity → Your balance is accurate
- Availability → You can access it anytime

🏢 Security Operations Center (SOC)

A **Security Operations Center (SOC)** is a centralized team responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity threats within an organization.

👉 It works **24/7** to ensure systems are secure.

◈ Purpose of SOC

- Detect cyber attacks early
- Prevent data breaches
- Monitor systems continuously
- Respond to incidents quickly

◈ SOC Workflow

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1. Data Collection

- Collect logs from:
  - Servers
  - Firewalls
  - Endpoints
  - Applications

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2. Monitoring

- Continuous observation of system activities
- Detect unusual behavior

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3. Detection

- Identify suspicious patterns
- Example:
  - Multiple failed logins
  - Unusual IP access

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4. Analysis

- SOC analyst investigates alerts
- Determines:
  - Real threat or false positive

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5. Response

- Take action:
  - Block IP
  - Isolate system
  - Remove malware

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6. Reporting

- Document incident
- Improve future detection

📊 Logs

Logs are **records of events** generated by systems, applications, and networks.

◈ Examples of Logs

- Login attempts
- File access
- System errors
- Network traffic

◈ Types of Logs

- **System Logs** → OS activities
- **Application Logs** → App behavior
- **Network Logs** → Traffic data
- **Security Logs** → Authentication events

◈ Why Logs are Important

- Detect suspicious activities
- Investigate incidents
- Provide evidence
- Track user actions

👀 Monitoring in SOC

Monitoring means continuously observing logs and system activities to identify threats.

◈ Types of Monitoring

- Real-time monitoring
- Continuous monitoring
- Automated monitoring

◈ Example

👉 If 100 failed login attempts occur:

- Monitoring system detects anomaly
- Generates alert

⚙️ SIEM (Security Information and Event Management)

◈ What is SIEM?

SIEM is a tool used in SOC to **collect, analyze, and correlate logs** from multiple sources.

◈ How SIEM Works

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1. Log Collection

- Collects data from:
  - Servers
  - Firewalls
  - Endpoints

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2. Normalization

- Converts logs into standard format

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3. Correlation

- Finds patterns across logs
- Example:
  - Multiple failed logins + unusual IP = attack

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4. Alert Generation

- Sends alerts for suspicious activity

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5. Visualization

- Dashboards for analysts

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6. Investigation

- Analysts analyze alerts

◈ Benefits of SIEM

- Centralized monitoring
- Faster detection
- Real-time alerts
- Better visibility

🚀 XDR (Extended Detection and Response)

◈ What is XDR?

XDR is an advanced security solution that provides **integrated threat detection across multiple layers**.

◈ What XDR Covers

- Endpoint (devices)
- Network
- Email
- Cloud

◈ How XDR Works

- Collects data from multiple sources
- Uses advanced analytics
- Detects complex attacks
- Provides automated investigation

◈ Advantages

- Reduces false positives
- Detects advanced threats
- Provides full visibility

🤖 SOAR (Security Orchestration, Automation and Response)

◈ What is SOAR?

SOAR is a platform that **automates security operations and incident response**.

◈ How SOAR Works

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1. Receives Alerts

- From SIEM or XDR

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2. Executes Playbooks

- Predefined workflows

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3. Automates Actions

- Block IP
- Isolate device
- Send alerts

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4. Reduces Manual Work

- Saves analyst time

◈ Benefits

- Faster response
- Automation
- Efficiency
- Reduced workload

🎯 Simple Example

👉 Scenario: Multiple failed logins from unknown IP

- **Logs** → Record login attempts
- **SIEM** → Detects unusual pattern
- **XDR** → Analyzes across systems
- **SOAR** → Automatically blocks IP

💻 Git Basics

Git is a distributed version control system used to track changes in files and manage projects efficiently. It helps in maintaining version history, collaboration, and backup of code.

📌 Key Concepts

- **Repository (Repo):**
  A folder that contains project files and Git tracking information.
- **Working Directory:**
  The area where files are created and modified.
- **Staging Area:**
  Temporary area where files are added before committing.
- **Commit:**
  A snapshot of changes saved in the repository.
- **Remote Repository:**
  An online repository (e.g., GitHub) used to store and share projects.

⚙️ Basic Git Commands

- **git init**
  Initializes a new Git repository.
- **git status**
  Displays the current status of files.
- **git add .**
  Adds all files to the staging area.
- **git commit -m "message"**
  Saves changes with a message.
- **git remote add origin <repo_link>**
  Connects local repository to GitHub.
- **git push -u origin main**
  Uploads files to the remote repository.

🔄 Git Workflow

1. Create project folder
2. Initialize repository (git init)
3. Add files (git add .)
4. Commit changes (git commit)
5. Connect to remote repository
6. Push to GitHub (git push)

🎯 Importance of Git

- Tracks changes in files
- Maintains version history
- Enables collaboration
- Prevents data loss